

Unraveling the effect of 600 keV carbon ion irradiation on the structural and magnetic properties of ZnO thin film

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HIGHLIGHTS

- Carbon beam induces stable and significant room temperature ferromagnetic ordering in ZnO thin films.
- Lighter ions prove to be better in controlled defect generation.
- Controlled damages result in ferromagnetic ordering in ZnO samples.
- Ion beam induced defects such as oxygen vacancy, zinc carbide (ZnC) etc., modulate the observed ferromagnetic ordering.
- Each carbon ions produces an average about 6.71 μ_B of the magnetic moment in ZnO samples.

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ABSTRACT

600 keV $^{12}\text{C}^+$ carbon ions have been irradiated upon 800 nm thin ZnO film. The thin ZnO films have been grown by pulsed laser deposition technique upon SiO_2/Si substrate. The glancing incidence X-ray diffraction study exhibits the hexagonal wurtzite crystal structure of ZnO, having space group $P6_3mc$. Systematic X-ray diffraction analysis of both unirradiated and irradiated samples, reveals that the strain decreases in the sample post irradiation. Room temperature magnetization studies have been performed using SQUID-VSM. As the 600 keV carbon ion fluence is varied, an onset of ferromagnetic ordering in the ZnO thin film is observed. The maximum average magnetic moment induced by each carbon ion is found to be about 6.71 μ_B . X-ray photoelectron spectroscopy has been utilized to analyze the oxidation state of the C/O/Zn, which also rules out the presence of any transition metal elements in carbon ions irradiated ZnO thin film samples. The chemical and structural analysis of the samples have been examined by Raman spectroscopy. It reveals a red shift in A_1 (LO) mode, which confirms the presence of tensile strain in the sample post irradiation. These experimental results, clearly indicate that oxygen vacancies and strain in the film trigger the onset of ferromagnetism in the ZnO thin film system.

1. Introduction

The discovery of Room Temperature Ferromagnetism (RTF) in Dilute Magnetic Semiconductor (DMS) materials has been growing very fast in past two decades, due to its practical application in spintronic devices [1–5]. Although the origin of ferromagnetism has been established in group III-V semiconductors, by doping with magnetic ions, however the maximum reported Curie temperature (T_C) in Mn-doped GaAs is only 173 K [6]. The quest for RTF in DMS has been a topic of interest in recent

years. The initial reports to produce DMS at room temperature have been with doping of Transition Metal (TM) elements into a semiconductor host material [7,8]. Amongst the several materials for DMS, ZnO and GaN have been extensively studied, since it has been theoretically predicted that these two materials are the potential candidates for DMS studies at room temperature [9]. This theoretical prediction opens a new path and voluminous experimental works have been carried out on various TM doped DMS systems [7,8,10–15]. The ferromagnetic ordering in ZnO has attracted the attention of scientific community

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across the globe, due to its direct wide band-gap (3.37 eV) and high exciton binding energy (60 meV) [16]. Xu et al. [17] have reported RTF in Mn-doped ZnO, and their results indicate that zinc vacancy (V_{Zn}^-) acceptor defects, lead to the observed magnetism. In addition to this, Yi et al. [18] have synthesized the Li-doped ZnO, using Pulsed Laser Deposition (PLD) technique and have reported that cation vacancies such as Li substitutional doping at cation site (Li_{Zn}) and Zn vacancies (V_{Zn}) are responsible for the observed ferromagnetism. In another report, Srivastava et al. [19] have reported RTF in Ni-ion implanted ZnO films, grown by PLD technique and the observed ferromagnetism has been pinned upon oxygen vacancies mediated Bound Magnetic Polaron (BMP) model.

On the other hand, studies also reveal that ferromagnetism in pure ZnO films can be produced by the zinc vacancies [20,21] and magnetic moments arise from unpaired 2p electrons at O site surrounding the zinc vacancy (V_{Zn}) defects [21]. Zhan et al. [22] have reported ferromagnetic ordering in the ZnO sample, due to substrate proximity effects. In this work, Zhan and coauthors have studied the room-temperature ferromagnetic ordering in pure ZnO thin films on silicon and quartz substrates to address the proximity effects of different substrates. They have suggested that RTF has been obtained from singly occupied oxygen vacancies created by thermal annealing. There is a significant enhancement of RTF in pure ZnO thin films on silicon substrate as compared to the quartz substrates, due to the interfacial reaction induced by silicon as oxygen getter, which produces more oxygen vacancies in the system and thereby enhances the observed RTF.

In recent years, not only transition metals but also non-magnetic ions have been doped in ZnO in the quest of observing RTF [23–35]. Pan et al. [23] have demonstrated experimentally that carbon-doped ZnO can be ferromagnetic at room temperature due to the presence of Zn–C complex structure in the host material. In their study, the carbon-doped ZnO thin films have been prepared by PLD technique, which exhibits RTF with $T_c > 400$ K. Li et al. [31] have also reported ferromagnetism in carbon-doped ZnO films prepared by PLD technique and the observed RTF has been explained by oxygen vacancies and interstitial site defects related magnetic correlations. However, the role of non-magnetic ions, in inducing ferromagnetism in ZnO material, is still a highly pursued field of research, due to good predictability and reproducibility of ion beam induced defects at optimum ion energy and fluence.

Ion beam induced ferromagnetism in carbon related systems viz. Highly Oriented Pyrolytic Graphite (HOPG), nanosized diamond film, fullerene etc., has been extensively studied experimentally as well as theoretically by various research groups [36–41]. Cervenka et al. [42] have elaborated that 2D point defects of certain kind (zig-zag edges) in HOPG system of high mosaicity, are responsible for intrinsic ferromagnetism due to the edge state of zig-zag edges. Ion beam irradiation technique has been established as a powerful tool for incorporating carbon in ZnO matrix for altering and assessing its properties [43–47]. Ion beam irradiation of appropriate energy, flux and fluence can create a wide range of defect density profile in the host matrix in a controlled manner and which could be responsible for the origin of ferromagnetism in wide band-gap semiconductors. Zhou et al. [43] have reported RTF in carbon implanted ZnO thin films, by varying the energy in the range of 8.8 keV–70 keV and by varying the fluence in the range of 9.84×10^{14} ions/cm² to 8.2×10^{15} ions/cm², where the origin of RTF has been explained on the basis of the chemical interaction of carbon with the host material. Kumar et al. [44] have reported ferromagnetism in 100 keV carbon ion implanted ZnO thin films, where authors have suggested that the observed ferromagnetic ordering is closely related to the distribution of carbon ions in the host matrix. In a recent work, Saravanan et al. [45] have also reported RTF in ZnO thin films, as a consequence of the 20 keV carbon ion irradiation upon ZnO thin films.

In the current manuscript, we present our work on the onset of ferromagnetism in ZnO thin films due to 600 keV carbon ion irradiation. The experimental analysis and discussion related to the finding have been reported in the subsequent sections in detail.

2. Experimental

ZnO thin films of size about 9 mm × 3 mm and thickness ~800 nm, have been grown from a ZnO ceramic target on SiO₂/Si substrate by pulsed laser deposition technique. The thin films have been irradiated at room temperature with ¹²C⁺ ions of 600 keV energy using 1.7 MV Tandetron particle accelerator (High Voltage Engineering, Europa). The carbon ions have been extracted from a high purity graphite powder filled SNICS (Source of Negative Ions by Cesium Sputtering) source. Two different fluences (1 pC/μm² and 10 pC/μm²) of ¹²C⁺ ions have been irradiated upon the ZnO thin films. The Glancing Incidence X-ray Diffraction (GIXRD) patterns of carbon ion irradiated ZnO and pristine samples have been acquired utilizing Bruker D8 Discover instrument with 4.5 kW Cu K_α rotating anode X-ray source of wavelength 1.5418 Å over the large Bragg angles in the range $30^\circ \leq 2\theta \leq 70^\circ$. The GIXRD measurement has been performed with step size 0.05° and scan rate 2° per minutes and with glancing angle of X-ray 0.4°. This study has been useful to confirm the crystal structure and phase purity of the samples, pre & post carbon ion irradiation. Room temperature magnetization measurement has been carried out using SQUID-VSM (Quantum design MPMS system) at UGC-DAE-CSR Kalpakkam Node. In order to understand the possible changes in the stoichiometry and to assess the involvement of any transition metal elements in carbon irradiated ZnO films, the samples have been investigated using X-ray photoelectron spectroscopy (VG ESCALAB MK200X). The XPS data has been acquired using a monochromatic X-ray source of Al K_α (1486.61 eV) having beam diameter of about 3 mm and collection area (with the largest slit) around (2 × 3) mm². The chemical changes and defects analysis of the samples have been assessed by Raman spectroscopy (Renishaw inVia, UK) with Ar⁺ ion laser source, having an excitation wavelength of 514 nm.

3. Results

3.1. Glancing incidence X-ray diffraction analysis

The Glancing Incidence X-ray Diffraction (GIXRD) data of pristine and carbon ion irradiated ZnO thin films taken at room temperature are shown in Fig. 1. Only (002) and (103) peaks of hexagonal wurtzite ZnO phase with space group *P6₃mc* have been observed. It is evident that the pristine film is highly crystalline and oriented along (103) plane. A

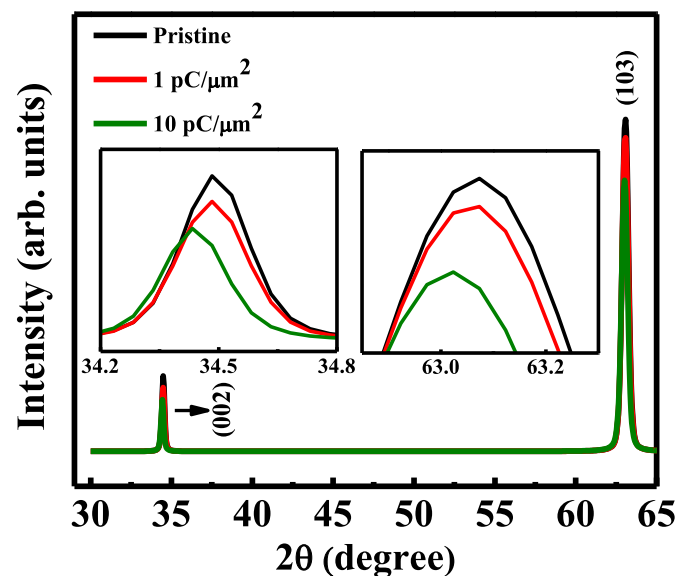


Fig. 1. GIXRD pattern of unirradiated and 600 keV carbon ion irradiated ZnO thin films at the fluences of 1 pC/μm² and 10 pC/μm². Insets depicting zoomed version of the corresponding (002) and (103) peaks, respectively.

relatively smaller but significant peak corresponding to c-axis orientation i.e., along (002) plane is also observed. The attenuation length of X-ray at an angle of incidence (0.4°) has been assessed to be approximately 152 nm. This value falls well within the thickness of the ZnO thin film, which is 800 nm. This fact is established by the GIXRD patterns in Fig. 1, which confirm the presence of (002) and (103) peaks originating from the ZnO thin film only. This is due to the lower attenuation length of X-ray (at glancing incidence of 0.4°) than the thickness of the film.

It is also evident from Fig. 1, that as a consequence of irradiation with the fluences of $1 \text{ pC}/\mu\text{m}^2$ and $10 \text{ pC}/\mu\text{m}^2$ of $^{12}\text{C}^+$ ions of energy 600 keV, the intensity of the peaks decreases and the corresponding peak position shifts towards a lower angle as compared to the pristine sample. This shift in peak position indicates the presence of tensile strain in the film as a consequence of carbon ion irradiation [48–50].

The crystallite size has been calculated for both peaks, i.e. (002) and (103), utilizing Debye-Scherrer formula [51],

$$\text{Crystallite size } (D) = \frac{(k\lambda)}{(\beta \cos \theta)} \quad (1)$$

where, k is the Scherrer's constant equal to 0.94, λ is the wavelength of the incident X-ray ($\lambda = 1.5418 \text{ \AA}$), β is the Full Width at Half Maximum (FWHM), and θ is the Bragg's angle. It can be observed from Fig. 2 (a) and (b) that there is a significant enhancement of crystallite size for both the observed planes i.e., (002) and (103), which indicates that although the intensity decreases as we increase the fluence, the crystallinity remains intact with the formation of grain and grain boundary in the crystal at higher fluence of carbon ion irradiation. It is also observed that the value of crystallite size is larger for the (002) plane as compared to the (103) plane. The average crystal size for (002) and (103) plane is 41.86 nm and 21.15 nm respectively.

Furthermore, the interplanar spacing (d) is estimated using Bragg's law equation [51],

$$n\lambda = 2d \sin \theta \quad (2)$$

where, n is the order of diffraction (usually $n = 1$), λ is the wavelength of the incident X-ray ($\lambda = 1.5418 \text{ \AA}$), and θ is the Bragg's angle.

For a hexagonal crystal structure, the interplanar spacing d is related to the lattice parameters a , c and the Miller indices h , k and l by the following relation [51],

$$\frac{1}{d^2} = \frac{4}{3} \left(\frac{h^2 + hk + k^2}{a^2} \right) + \frac{l^2}{c^2} \quad (3)$$

The lattice parameter ' c ' is determined using the equation [51],

$$c = \lambda / \sin \theta \quad (4)$$

where, λ is the X-ray wavelength and θ is the Bragg diffraction angle corresponding to (002) peak. The calculated lattice constants ($a = b$ & c) and interplanar spacing (d) values from GIXRD data are found to be in good agreement with standard values of JCPDS card no. 36–1451 and

are tabulated in Table 1. It is observed from Fig. S1 (a) and (b) (supplementary data file) that at the fluence of $1 \text{ pC}/\mu\text{m}^2$ irradiation, the value of interplanar spacing is found to be minimum. It has also been noticed that at the fluence of $1 \text{ pC}/\mu\text{m}^2$, the lattice parameter ' c ' decreases, while lattice parameter ' a ' increases, as compared to the pristine sample as shown in Fig. S1 (c) and (d) (supplementary data file). The decrease in the lattice parameter ' c ' is due to the formation of structural vacancies in the sample post carbon ion irradiation [52].

The lattice strain present in the thin films has been calculated using the following Wilson equation [53],

$$\epsilon_w = \frac{(\beta \cot \theta)}{4} \quad (5)$$

The strain values decrease monotonically as we increase the ion fluence for both the observed (002) and (103) planes, as shown in Fig. 3 (a) and (b) respectively. The range of strain values for (002) plane, is $(3.087\text{--}2.718) \times 10^{-3}$ and that for (103) plane is $(3.309\text{--}3.220) \times 10^{-3}$. The decrease in strain value for (002) plane is higher than that of (103) plane.

The dislocation density (δ), which is the measure of dislocation lines per unit volume in the sample, is determined using the following equation [51],

$$\text{Dislocation Density } (\delta) = \frac{1}{D^2} \quad (6)$$

where, D is the crystallite size. The dislocation density is found to decrease upon successive irradiation as shown in Fig. 3 (c) and (d).

In case of highly c-axis oriented hexagonal crystals, the in plane stress (σ) has been calculated, based upon the biaxial strain model [54] given by,

$$\sigma = 4.5 \times 10^{11} [(c_0 - c)/c_0] \text{ N/m}^2 \quad (7)$$

where, c is the lattice parameter of the (002) reflection in the GIXRD spectra and c_0 is the corresponding bulk value ($5.2066 \text{ \AA}$). The unirradiated and $1 \text{ pC}/\mu\text{m}^2$ of fluence irradiated samples have tensile stress for (002) orientation, which turns to compressive stress at the higher fluence of $10 \text{ pC}/\mu\text{m}^2$, as shown in Fig. S2 (supplementary data file).

3.2. SRIM analysis

We have carried out Monte Carlo based Stopping Range of Ions in Matter (SRIM) simulation [55,56] for 600 keV carbon ions in ZnO as shown in Fig. 4 (a). We have calculated the range of the 600 keV carbon ions into ZnO using SRIM code, which is found to be about 790 nm and thickness of our thin film is 800 nm [55,56]. Therefore, we have selected 600 keV ion energy, so that the carbon ions remain limited to the ZnO thin films and create the defects only inside the thin films. It is also found that the maximum vacancies have been created at the depth of 744 nm as per SRIM calculation. This way there will be no damage to the SiO_2/Si substrate. It has been simulated that 600 keV carbon ions produce an

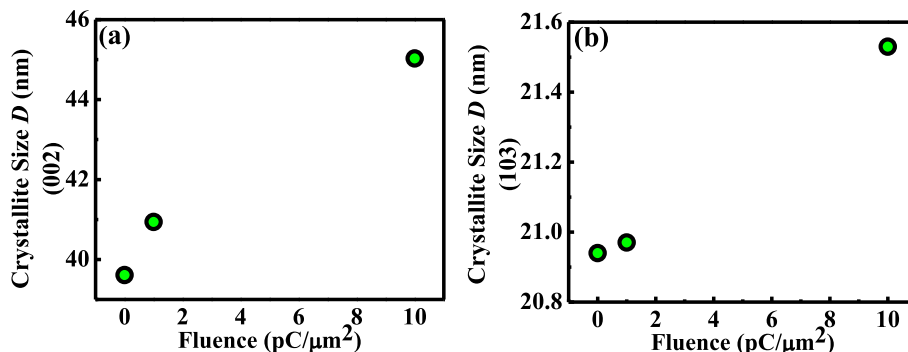


Fig. 2. (a) & (b) Crystallite size vs. 600 keV carbon ion fluence for (002) and (103) planes respectively.

Table 1

Estimated inter-planar spacing, lattice parameter, crystallite size, stress, lattice strain and dislocation density from GIXRD data for (002) & (103) planes of pristine sample, 1 pC/ μm^2 and 10 pC/ μm^2 fluence irradiated samples.

Samples	$d(002)$ (Å)	$d(103)$ (Å)	a (Å)	c (Å)	D (002) (nm)	D (103) (nm)	σ (002) (GPa)	ϵ (002)	ϵ (103)	δ (002) lines/ m^2	δ (103) lines/ m^2
Standard values	2.60330	1.4771	3.2498	5.2066	–	–	–	–	–	–	–
Pristine	2.60152	1.4743	3.2322	5.20304	39.61	20.94	0.31	3.087×10^{-3}	3.309×10^{-3}	6.374×10^{14}	22.806×10^{14}
Irradiated with 1 pC/ μm^2	2.60088	1.4742	3.2335	5.20176	40.94	20.97	0.42	2.986×10^{-3}	3.303×10^{-3}	5.966×10^{14}	22.741×10^{14}
Irradiated with 10 pC/ μm^2	2.60421	1.47498	3.22893	5.20842	45.03	21.53	−0.16	2.718×10^{-3}	3.220×10^{-3}	4.932×10^{14}	21.573×10^{14}

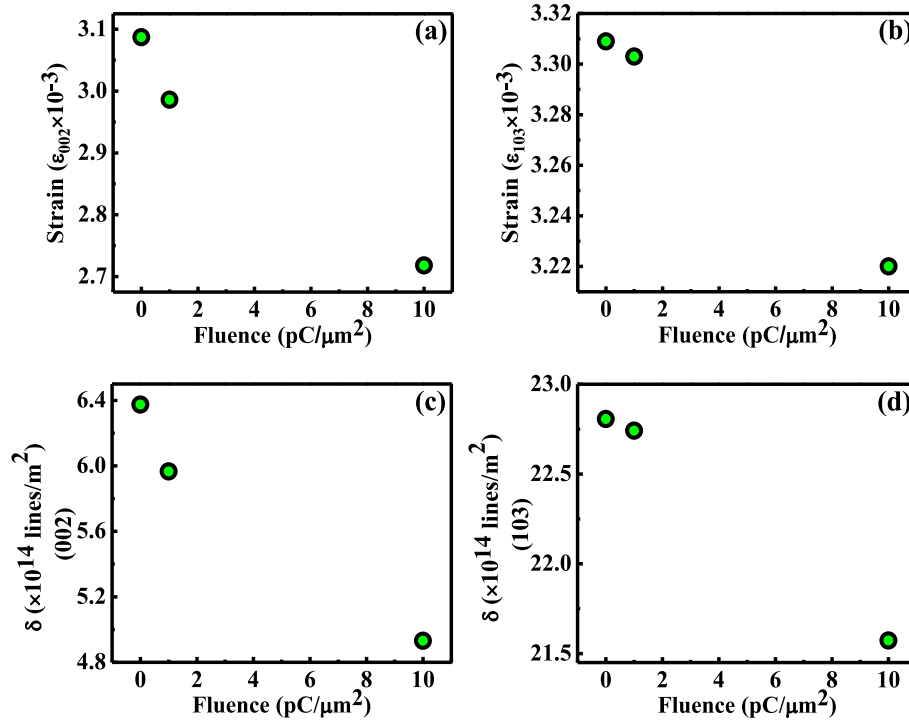


Fig. 3. (a) & (b) Strain vs. 600 keV carbon ion fluence for (002) and (103) planes respectively. (c) & (d) Dislocation density vs. 600 keV carbon ion fluence for (002) and (103) planes respectively.

average of about 720 vacancies per ion in the ZnO thin films as per SRIM code [55,56].

It can be noticed from Fig. 4 (a) that the carbon deposition depth profile for 600 keV carbon ion irradiation in ZnO, has a maximum at the target depth of 768 nm, which also indicates that the maximum vacancy density created by carbon ion lies near the end of the ion range. Fig. 4 (b) and (c) illustrate the distribution of carbon concentration along the target depth as a result of 600 keV carbon ion irradiation in the ZnO thin film, corresponding to the ion-fluences of 1 pC/ μm^2 and 10 pC/ μm^2 respectively. This illustrates that the distribution of carbon ions in the ZnO matrix is nonlinear and reaches its maximum at the end of the ion range. It can be observed from Fig. 4 (b) that the concentration of carbon in ZnO system is quite large at the higher ion-fluence compared to that of at the lower fluence. At higher fluence, due to the presence of a large number of carbon ions in ZnO matrix, it is expected that a non-uniform distribution of carbon ions establishes in the sample which leads to the decrease in magnetization value.

3.3. SQUID analysis

The room temperature magnetization measurements of both unirradiated and 600 keV carbon ion irradiated ZnO thin films have been

shown in Fig. 5. The study of magnetic ordering has been performed using SQUID-VSM. The fluences of 600 keV carbon-ion irradiations have been 1 pC/ μm^2 and 10 pC/ μm^2 . The unirradiated sample reveals, a very strong diamagnetic nature with negative slope. The sample shows, a promising response to the ion beam irradiation with a fluence of 1 pC/ μm^2 . The advent of small ferromagnetic ordering in ZnO thin film as a consequence of the carbon-ion irradiation, can be observed from Fig. 5. The small ferromagnetic kink, post irradiation has been observed and the sample still showed a large diamagnetic behavior of the substrate.

The magnetization data for unirradiated and irradiated samples post subtraction of the diamagnetic background of the sample and substrate, are shown in Fig. 6. It can be clearly seen in Fig. 6, that at lower fluence of 1 pC/ μm^2 , the observed saturation magnetization is higher, which significantly decreases with increase in fluence further to 10 pC/ μm^2 . However, it can also be noticed that with the increase in the ion fluence, the highly crystalline structure of ZnO gets altered mildly. This is supported by GIXRD data of irradiated and unirradiated samples under similar conditions.

Although the GIXRD peaks of irradiated and unirradiated samples remain well defined and sharp, but the intensity of peaks decrease monotonically as the fluence is increased. This means that, at higher fluence of 10 pC/ μm^2 , the defect density increases non-uniformly within

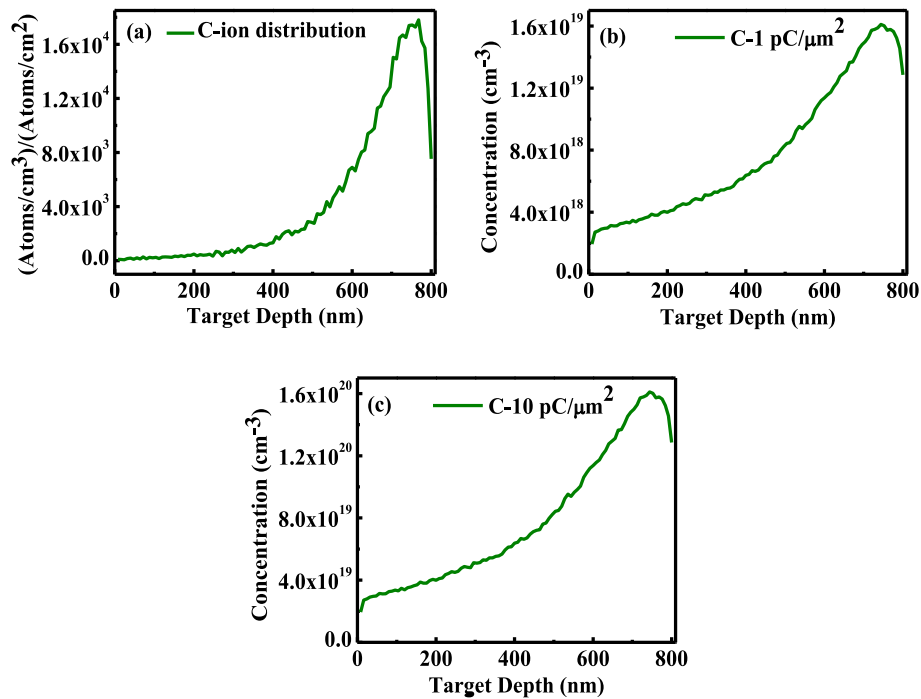


Fig. 4. SRIM-TRIM simulated (a) carbon deposition depth profile with respect to target depth, (b) carbon concentration for the fluence of $1 \text{ pC}/\mu\text{m}^2$ & (c) carbon concentration for the fluence of $10 \text{ pC}/\mu\text{m}^2$ as a function of target depth, for 600 keV carbon ion irradiated ZnO thin film.

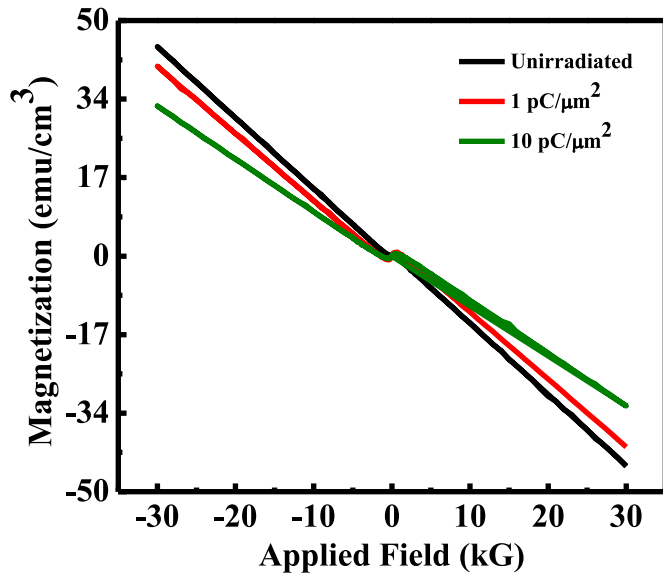


Fig. 5. Magnetization vs applied field measurements of unirradiated and carbon ion irradiated ZnO samples without subtraction of diamagnetic contribution.

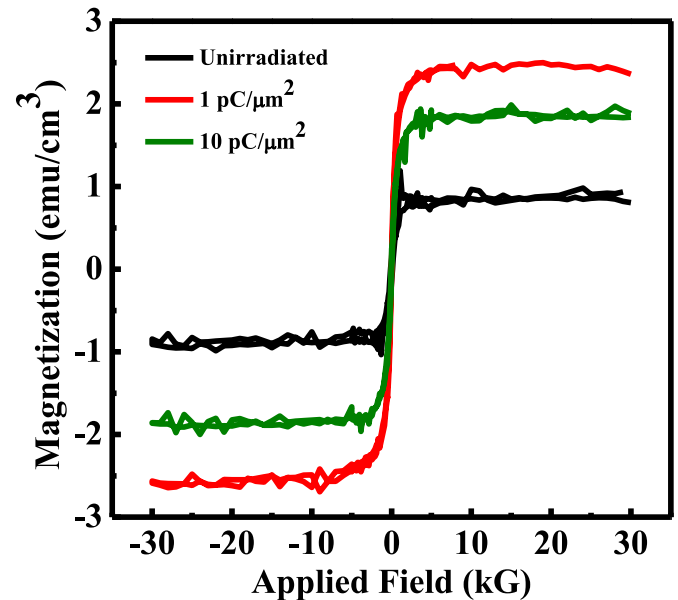


Fig. 6. Magnetization vs applied field measurements of unirradiated and carbon ion irradiated ZnO samples after subtracting the diamagnetic contribution.

the matrix of ZnO and this forces carbon atoms to occupy some interstitial positions, which leads to the decrease in saturation magnetization at higher fluence of $10 \text{ pC}/\mu\text{m}^2$ [34,43,44]. We can see the dependence of saturation magnetization on the irradiation fluence. At lower fluence of $1 \text{ pC}/\mu\text{m}^2$, the defects created by carbon ion irradiation have been uniformly distributed in the system and the defect density created by carbon ion is not detrimental to the observed magnetic ordering. A similar study by Zhou et al. [43] have reported that the highest saturation magnetic moment is observed for as low as 1 at. % carbon content, whereas higher carbon concentration (5 at. %) results in lower saturation magnetic moment. Thus, at higher fluence the increased defect

density is responsible for the decrement of the magnetic ordering. We have estimated the average magnetic moment induced by each carbon ion, and it is found that each carbon ion produces an average of about $6.71 \mu_B$ of magnetic moment in ZnO matrix, at $1 \text{ pC}/\mu\text{m}^2$ fluence of carbon implantation in ZnO. Thus, the room temperature ferromagnetism is clearly observed in both the samples as a consequence of carbon ion irradiation. Therefore, the above discussion suggests that the intrinsic defects created by carbon ion irradiation with milder defect density, is responsible for the better magnetic ordering in the system.

3.4. Raman spectroscopy analysis

Raman spectroscopy is a versatile tool to understand the irradiation effect of ions, viz., defects and structural disorder in the host matrix. Fig. 7 depicts the room-temperature Raman spectra of both pristine as well as carbon ion irradiated ZnO thin film (fluence 1 pC/ μm^2) ranging from 400 cm^{-1} to 650 cm^{-1} . It is observed from Fig. 7, that the pure and the carbon ion irradiated ZnO thin film samples, exhibit hexagonal wurtzite structure having space group C_{6v}^4 ($P6_3mc$) [57,58]. The optical phonons at the Γ point of the Brillouin zone with two formula units in the primitive cell denoted by the following relation [57].

$$\Gamma_{\text{opt}} = A_1 + E_1 + 2E_2 + 2B_1$$

The fact that the A_1 and E_1 modes are polar, indicates that these modes split into transverse-optical (TO) and longitudinal-optical (LO) phonons. These modes are Raman and infrared active. The E_2 mode ascribed as nonpolar, splits into two wave numbers referred as E_2 (high) and E_2 (low), which are associated with the motion of oxygen and zinc sublattices respectively. The nature of only these two nonpolar E_2 modes i.e., E_2 (high) and E_2 (low) remain Raman active. In Raman scattering, the B_1 modes are infrared and Raman inactive, hence they are referred to as silent modes.

The Raman spectra of the pristine ZnO film exhibit peaks at 436 cm^{-1} and 578 cm^{-1} as shown in Fig. 7, which correspond to the vibrational modes E_2 (high) and A_1 (LO) respectively [59]. The low-intense peak at 436 cm^{-1} is due to vibration of oxygen sublattice, which is the characteristic of hexagonal wurtzite ZnO crystal structure. The high-intense peak at 578 cm^{-1} arises due to O and Zn vacancies/interstitials type of defects or their complexes [58,60,61], which is consistent with our GIXRD results.

In Fig. 7, a very high intense peak at 521 cm^{-1} indicates the presence of the LO Raman mode due to scattering from Si substrate. The attributes and details of the Raman peaks are described in Table 2. It is also observed that even after irradiation, E_2 (high) mode remains intense, while defect-induced A_1 (LO) mode becomes more intense, which confirms that the origin of room temperature ferromagnetism in our case is attributed to the structural disorder and defects created by carbon ion irradiation. It is worth noticing from Table 2, that post irradiation with fluence of 1 pC/ μm^2 , the peak position of E_2 (high) and defect-induced A_1 (LO) modes shift towards higher and lower wave number respectively, in comparison to pure ZnO peaks. These shifts in peak position of

Table 2

Phonon modes of unirradiated and carbon ion irradiated hexagonal wurtzite ZnO nanostructure.

Vibration frequency (cm^{-1})		Assignment
Unirradiated ZnO	Irradiated ZnO (1 pC/ μm^2)	
436	439	E_2 (high)
521	521	LO Raman mode of Si
578	576	A_1 (LO)

E_2 (high) and A_1 (LO) modes are referred as blue shift and red shift respectively. The shift in E_2 (high) mode indicates the fact that C^+ substitution creates the structural changes in the host ZnO lattice and disruption of its translational symmetry without changing the original structure, while the shift in A_1 (LO) mode reveals the presence of tensile strain in the sample.

3.5. XPS analysis

A systematic XPS study is carried out to deduce the chemical bonding states of atoms, impurities present in the carbon ion irradiated ZnO thin film, and to rule out the presence of transition metal elements if any. Fig. 8 (a) represent the C 1s and 8 (b) indicate the O 1s spectrum, corresponding to the sample irradiated with 10 pC/ μm^2 of carbon ion fluence along with their fitted curve.

The XPS spectra is fitted with Voigt functions. The binding energy scale was measured after charge correction with the help of C 1s peak at 284.6 eV. The XPS spectrum of carbon shows an asymmetric 1s peak in Fig. 8 (a), which is de-convoluted into two peaks. The high intense peak at lower binding energy of 284.6 eV is associated with the presence of free carbon in the chamber, while low intense peak at higher binding energy of 286.6 eV is attributed to the zinc oxy-carbide complex structure within ZnO matrix, which reveals the carbon substitution leading to the formation of Zn–C bonds in the system [23,44]. It can be observed from Fig. 8 (b), that there are the two peaks located at binding energies of 529.8 eV and 532.4 eV. The peak at low binding energy of 529.8 eV can be attributed to Zn–O bonding state or presence of O^{2-} ions in the hexagonal wurtzite structure of ZnO bound to Zn^{2+} ions [26,29], whereas high intense peak at higher binding energy of 532.4 eV is ascribed to oxygen vacancies, absorbed oxygen species or hydroxyl groups [31,46,62]. It is also noticed that oxygen species at higher binding energy, indicates that the oxygen ions appear in the oxygen-deficient regions, within the host ZnO. This validates the presence of oxygen vacancies in the system, which enhance the saturation magnetization in carbon ion irradiated ZnO thin films [44]. Thus, the XPS results, suggest the carbon substitution for oxygen and in addition to this there is also a possibility of such defect complexes that involve formation of Zn–C bonds/complexes. The XPS results also rule out the presence of transition metal impurities on the cleaned surface of the ZnO thin film.

4. Discussion

There are various aspects for the origin of ferromagnetism in DMS materials. There are two main questions which require our attention to explain the observed magnetic ordering; one is to recognize the primary source of magnetization and the other is to rule out the presence of any transition metal impurity in the sample. In fact, a lot of literature is available related to the origin of magnetic ordering in ZnO system, by doping/irradiation with transition metals, non-magnetic elements and rare earth elements, from both experimental and theoretical point of view [63–71]. Some of these reports have been explained by formation of magnetic cluster/secondary phases or by TM doping in oxides or semiconductors resulting in various kinds of defects such as oxygen vacancy (V_o), zinc vacancy (V_{Zn}), zinc interstitial (Zn_i) etc., leading to RTF [72–75].

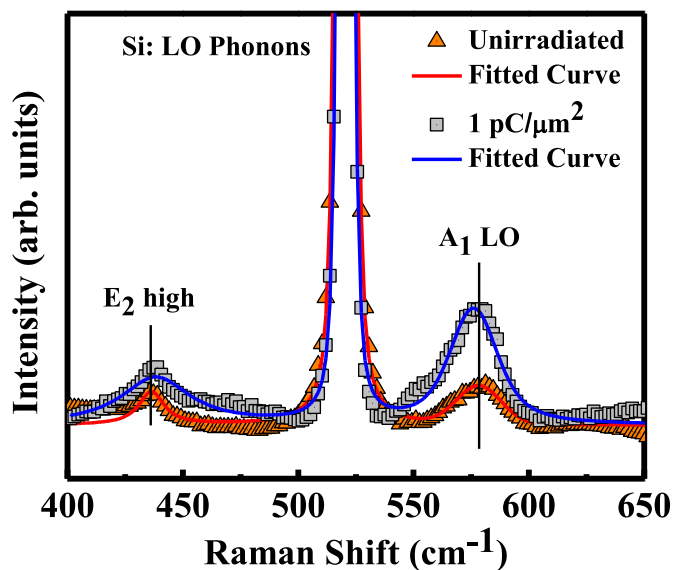


Fig. 7. Raman Spectra of unirradiated ZnO and 600 keV carbon ion irradiated ZnO thin film sample with the fluence of 1 pC/ μm^2 .

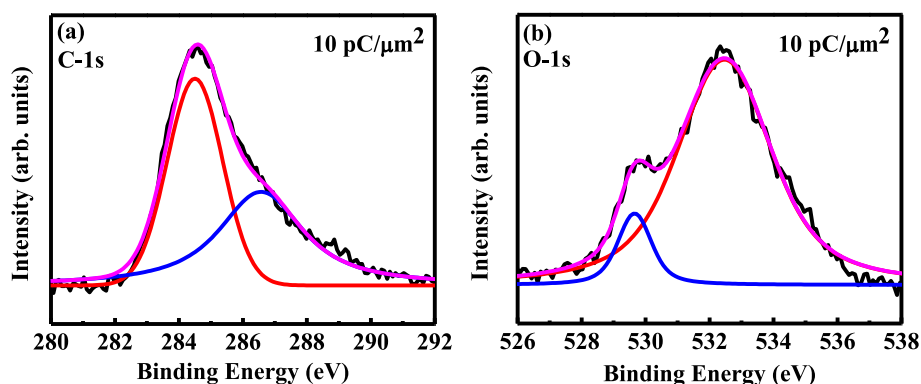


Fig. 8. X-ray photoelectron spectra of carbon ion irradiated ZnO thin film sample with the fluence of $10 \text{ pC}/\mu\text{m}^2$: (a) C-1s core-level spectra (b) O-1s core-level spectra.

Borges et al. [76] have reported magnetic ordering in ZnO single crystal, doped with Co ion implantation, where “as-implanted” sample shows superparamagnetic behavior and vacuum annealed sample at 800°C , indicates RTF. This behavior is attributed to the formation of Co clusters in as-implanted sample and the formation of larger Co clusters of dimension 15 nm in the samples after annealing in vacuum. Furthermore, Zhou et al. [77] also studied the structural and magnetic properties of Co and Ni implanted ZnO single crystals. They have reported the formation of Co and Ni nanocrystals in as-implanted samples and these nanocrystals trigger the superparamagnetic behavior in the system. In contrast to this, Zhou et al. [78] have reported defect induced ferromagnetism in vacuum annealed pristine ZnO crystals as well as Fe implanted ZnO crystals. They have concluded that the observed magnetization in vacuum annealed ZnO and Fe implanted ZnO crystals is related to the defect, though the formation of metallic secondary phases are suppressed. In further work, Zhou et al. [79] have studied the transition metal (Fe, Ni, Co) induced ferromagnetism in ZnO single crystals. However, they have not observed any indication for ferromagnetic DMS, created by ion implantation, while they have reported that the formation of metallic secondary phases of Fe, Ni and Co nanocrystals occur post ion implantation in the system and these secondary phases are primary source of ferromagnetism. Zhang et al. [80] have reported the onset/reduction of ferromagnetic ordering in n-type ZnO film at 10 K , as a consequence of varying Fe^+ fluence and subsequent annealing at 700°C for 1 h . The reduction in the observed ferromagnetic ordering has been explained by the enhanced anti-ferromagnetic coupling, due to higher oxygen vacancy density at higher fluences of implantation. In another work, Zhang et al. [81] have reported that the observation of RTF in Fe^+ ions implanted ZnO films would be mediated by Fe substituted at Zn site acting as acceptor and this leads to the localized spins under similar conditions as mentioned in above reference such as same ion, energy, fluences and synthesis process.

Sanyal et al. [82] have reported the RTF in ZnO pellets, as a consequence of 1.63 MeV , $^{56}\text{Fe}^{6+}$ ions ($1 \text{ at. \%}$ doping) implantation followed by annealing at 500°C . They have not reported any formation of secondary phases or formation of nanocrystals, post Fe implantation. However, they have found that unannealed Fe implanted samples do not reveal any signature of ferromagnetism, while annealed samples exhibit very high ferromagnetic ordering as compared to chemically doped ZnO samples. In addition to this, they have also discussed, why chemical doping of Fe ($\sim 1\text{--}2 \text{ at. \%}$) in ZnO does not yield the same level of ferromagnetism, as can be obtained at the same concentration by direct Fe ion beam implantation. They have suggested that ion implantation helps to create large number of Zn vacancies from lattice sites and during annealing process, these vacant sites are replaced by Fe instead of Zn, due to higher valency of Fe as compared to Zn, leading to significant enhancement of ferromagnetism, while in chemically doped unannealed ZnO pellets, the Zn vacancy density is much smaller, which may be

responsible for suppression of the ferromagnetic ordering. It has been reported by Pandey et al. [83], that exchange interaction between charge carriers created by oxygen vacancies and localized spins of Ni ions is the main source of RTF in 200 keV , Ni^{2+} ion implanted ZnO thin films (thickness $\sim 130 \text{ nm}$). In another work, Pandey et al. [84] have reported that the onset of RTF in 200 keV , Ni^{2+} ion implanted ZnO thin films, having mostly same fluences and thickness $\sim 800 \text{ nm}$, is due to defect mediated mechanism related to BMP model. Similarly, Srivastava et al. [19] have also proposed from their experimental studies, that larger number of defects such as oxygen vacancies, closely associated with BMP model, lead to higher saturation magnetization at fluence of $6 \times 10^{15} \text{ ions}/\text{cm}^2$ in 200 keV , Ni^{2+} ion implanted ZnO thin films.

Satyarthi et al. [59] have studied the RTF in 500 keV Xe^{3+} ions irradiated ZnO single crystals and thin films having thicknesses $\sim 500 \mu\text{m}$ and $\sim 435 \text{ nm}$ respectively at different fluences ($1 \times 10^{16} \text{ ions}/\text{cm}^2$, $5 \times 10^{16} \text{ ions}/\text{cm}^2$, $2 \times 10^{17} \text{ ions}/\text{cm}^2$, and $3.5 \times 10^{17} \text{ ions}/\text{cm}^2$). However, they did not observe any RTF in pristine as well as irradiated ZnO single crystals, while pristine and irradiated ZnO thin films show RTF with significant enhancement of saturation magnetization up to $2 \times 10^{17} \text{ ions}/\text{cm}^2$ of the fluence. They have suggested from their experimental studies, that the density of oxygen vacancies is responsible for the absence and presence of RTF in ZnO single crystals and thin films respectively. Hariwal et al. [49] have studied the RTF in 60 keV , N ion implanted ZnO thin films, having thickness $\sim 300 \text{ nm}$, by varying angle of implantation at fluence of $1 \times 10^{16} \text{ ions}/\text{cm}^2$. It has been suggested that RTF in N ion implanted ZnO thin films, is a result of formation of oxygen vacancies, grains and grain boundaries. The incorporation of N ions into ZnO films induces magnetic moment's interaction between N and O ions due to p-p interaction in the host matrix. It has also been predicted that the higher density of V_o is also a key factor for significant enhancement of saturation magnetization because higher density of V_o creates a higher order of unpaired states in the system. Pan et al. [23] have reported that formation of Zn-C system produces RTF in carbon doped ZnO semiconductor, suggesting that only carbon in carbide form, contributed to the magnetic moment and not the other induced defects. Their theoretical calculation showed that coupling between $\text{C}2\text{p}$ and $\text{O}2\text{p}$ localized spins by a p-p interaction, which arises from carbon substitution at oxygen site (C_o), creates holes in $\text{O}2\text{p}$ states, that pushes hole-mediated spin alignment of parent carbon atoms, leading to indirect ferromagnetic coupling of carbon atoms. This behavior of p-p interaction in carbon doped ZnO, is quite similar to p-d hybridization in TM doped semiconductors. Their experimentally reported magnetic moment values for $1 \text{ at. \%}$ and $5 \text{ at. \%}$ carbon doped ZnO films, are in the range of $(2.5\text{--}3.0) \mu_\text{B}/\text{C}$ and $(1.5\text{--}2.5) \mu_\text{B}/\text{C}$ respectively, which is in good agreement with their theoretical prediction of $2.02 \mu_\text{B}/\text{C}$ of magnetization [23]. Zhou et al. [43] have also proposed RTF in carbon-implanted ZnO films. They found that uniform distribution of carbon ions in ZnO, induce the ferromagnetism and not the defects

created by carbon ion implantation, which further decreases at higher carbon concentrations. Their highest estimated saturation magnetic moment value is $0.24 \mu_B/\text{C}$ for 1 at. % carbon implanted ZnO at 5 K. Similarly, Kumar et al. [44] have also suggested that ferromagnetism in carbon implanted ZnO thin films, is strongly influenced by the distribution of carbon ions within the host lattice. Apart from this, Wei et al. [85] have fabricated carbon doped ZnO nanoneedles by Ar^+ ion irradiation. They have suggested that dipole-dipole interaction amongst nanoneedles, leads to the formation of ferromagnetic anisotropy in the system. Furthermore, Wang et al. [46] have proposed from both the experimental and theoretical studies, that primarily formation of carbon interstitial (C_i) defects, took place in carbon implanted ZnO nanowires. They have experimentally reported that saturation magnetization in carbon implanted ZnO, is 10 times higher than that of pristine ZnO nanowires. The $\text{C } 2p(\pi^*)$ derived states of residual carbon interstitial (C_i) defects, is primarily responsible for the origin of the enhancement of RTF in the bulk region. According to Gautam et al. [47] carbon implanted ions in ZnO thin films, form ZnCO_3 complex phase at interstitial position by replacing some oxygen sites with carbon atoms, which are mainly responsible for the RTF. A similar study has been carried out by Saravana et al. [45], in which they have reported RTF as a consequence of 20 keV carbon ion implanted ZnO thin film. Their results suggested that charge transfer between O-2p and carbon interstitial (C_i) defects is mainly responsible for the ferromagnetism at lower fluence. At higher fluence, they have reported that implanted carbon ions replaces the lattice and creates Zn vacancies in the system, which leads to the ferromagnetism in ZnO thin film. In addition to this, they have also studied the effect of annealing on magnetism. They found that after annealing at 500°C , the magnetization of low fluence irradiated sample decreases, due to annealing of carbon interstitial (C_i) defects, while the magnetization of high fluence sample remains almost same due to presence of highly stable defects in the system.

It is observed from Table 3 that there is a significant variation in the saturation magnetization values, obtained by various research groups at various irradiation fluences (at different energies) as well as by various chemical doping concentrations. In the present work, the saturation magnetization at the fluence of $1 \text{ pC}/\mu\text{m}^2$ is almost three times higher than that of the pristine sample, which follows a similar pattern as reported by Kumar et al. [44]. The estimated magnetic moment induced by each carbon ion, corresponding to the sample irradiated with the fluence of $1 \text{ pC}/\mu\text{m}^2$ is $6.71 \mu_B$, which is significant and maximum compared to previously reported values as per Table 3. The significant enhancement in the magnetic moment induced by each carbon ion at the

fluence of $1 \text{ pC}/\mu\text{m}^2$ may be due to the presence of uniform and stable oxygen vacancies in the ZnO system.

Apart from defects such as oxygen vacancy (V_O), zinc vacancy (V_Zn), zinc interstitial (Zn_i), formation of magnetic cluster/secondary phases etc., various groups have shown strain induced ferromagnetism. Liu et al. [50] have studied strain induced ferromagnetic ordering in Co-doped ZnO thin films, grown by radio-frequency magnetron sputtering. In this study, they have suggested that strain increases the Co–O bond length and the amount of defects in the system, when film thickness is thinnest (15 nm), and this results in significant enhancement of room temperature ferromagnetism in Co-doped ZnO thin film. Recently, Kumar et al. [86] have studied the strain induced structural and magnetic properties in thin MoS_2 flakes upon 1.5 MeV proton ion irradiation. They have suggested that increment of strain induced the amount of defects in the sample in the form of sulfur vacancies, which lead to the enhancement of saturation magnetic moment as ion fluences is increased.

In the present study, we find that the defect as well as the strain play an important role in the structural and magnetic properties of 600 keV carbon ion irradiation upon ZnO thin film. In GIXRD analysis of the ZnO samples, the shift in peak positions towards lower angle, is a clear indication of the presence of tensile strain in ZnO samples, post carbon ion irradiation. Although, we can see the decrement in the lattice strain post irradiation, but the values remain positive for pristine sample as well as $1 \text{ pC}/\mu\text{m}^2$ and $10 \text{ pC}/\mu\text{m}^2$ of fluence irradiated samples (Table 1). At the fluence of $10 \text{ pC}/\mu\text{m}^2$, the intensity of the GIXRD peak decreases and also shifts towards lower angle as compared to the pristine sample and $1 \text{ pC}/\mu\text{m}^2$ of fluence irradiated ZnO sample. This shift indicates decrease in lattice strain, stress and amount of defect density (dislocation density) as observed in Table 1. This decrement in the values of lattice strain, stress and dislocation density, indicates that the defects are non-uniformly distributed in the ZnO host matrix, which leads to reduction in saturation magnetization at the fluence of $10 \text{ pC}/\mu\text{m}^2$. The Raman data also corroborate well with the GIXRD data as defect induced peak at 578 cm^{-1} is also shifted towards lower vibration frequency post carbon ion irradiation, which also indicates the presence of tensile strain in the sample. In VSM measurement, we have observed the value of induced magnetic moment to be $6.71 \mu_B/\text{C}$ ion for the fluence of $1 \text{ pC}/\mu\text{m}^2$, which is significant as compared to previously reported values [23, 43]. This increase in value of induced magnetic moment per carbon ion can be related to the strain and defect induced by carbon ion irradiation in ZnO host matrix, which is well supported by our GIXRD and Raman results. The defects created by carbon ion irradiation at the fluence of 1

Table 3

The comparison chart of saturation magnetization (M_s) and the magnetic moment (μ_B) per C for carbon ion irradiated ZnO and similar systems at room temperature.

Sample and nature	Conditioning	Fluence	M_s	μ_B/C	Reference
ZnO thin film	100 keV C ions implantation	Pristine	$1.7 \text{ emu}/\text{cm}^3$	–	Kumar et al. [44]
		$5 \times 10^{16} \text{ ions}/\text{cm}^2$	$6.2 \text{ emu}/\text{cm}^3$	–	
		$1 \times 10^{17} \text{ ions}/\text{cm}^2$	$4.6 \text{ emu}/\text{cm}^3$	–	
ZnO thin film	20 keV C ions implantation	Pristine	–	–	Saravanan et al. [45]
		$2 \times 10^{16} \text{ ions}/\text{cm}^2$	$6.0 \text{ emu}/\text{cm}^3$	–	
		$4 \times 10^{16} \text{ ions}/\text{cm}^2$	$5.0 \text{ emu}/\text{cm}^3$	–	
		$6 \times 10^{16} \text{ ions}/\text{cm}^2$	$6.5 \text{ emu}/\text{cm}^3$	–	
ZnO-nanowires	40 keV $^{12}\text{C}^+$ ions implantation	Pristine	$0.3 \text{ emu}/\text{cm}^3$	–	Wang et al. [46]
		$1.5 \times 10^{16} \text{ ions}/\text{cm}^2$	$2.7 \text{ emu}/\text{cm}^3$	–	
ZnO thin film	40 keV $^{12}\text{C}^+$ ions implantation	Pristine	–	–	Gautam et al. [47]
		$1.5 \times 10^{16} \text{ ions}/\text{cm}^2$	$5.4 \times 10^{-4} \text{ emu}$	–	
ZnO thin film	0 at. % of C-doped	–	$0 \text{ emu}/\text{cm}^3$	–	Pan et al. [23]
	1 at. % of C-doped	–	$5.1 \text{ emu}/\text{cm}^3$	$(2.0-3.0) \mu_B/\text{C}$	
	5 at. % of C-doped	–	$7.1 \text{ emu}/\text{cm}^3$	$(1.5-2.5) \mu_B/\text{C}$	
ZnO powder	Pristine ZnO	–	–	–	Ye et al. [33]
	5 mol % of C-doped	–	$9.3 \times 10^{-4} \text{ emu}/\text{g}$	$0.0153 \mu_B/\text{C}$	
	10 mol % of C-doped	–	$11.8 \times 10^{-4} \text{ emu}/\text{g}$	$0.0123 \mu_B/\text{C}$	
	15 mol % of C-doped	–	$15.2 \times 10^{-4} \text{ emu}/\text{g}$	$0.0098 \mu_B/\text{C}$	
ZnO thin film	600 keV C ions irradiation	Pristine	$0.85 \text{ emu}/\text{cm}^3$ or $(0.15 \text{ emu}/\text{g})$	–	Present Work
		$1 \text{ pC}/\mu\text{m}^2$ or $(6.25 \times 10^{14} \text{ ions}/\text{cm}^2)$	$2.43 \text{ emu}/\text{cm}^3$ or $(0.43 \text{ emu}/\text{g})$	$6.71 \mu_B/\text{C}$	
		$10 \text{ pC}/\mu\text{m}^2$ or $(6.25 \times 10^{15} \text{ ions}/\text{cm}^2)$	$1.85 \text{ emu}/\text{cm}^3$ or $(0.33 \text{ emu}/\text{g})$	$0.51 \mu_B/\text{C}$	

$\text{pC}/\mu\text{m}^2$ are uniformly distributed in the system, which result in significant enhancement of saturation magnetization at this fluence. We have also observed defect induced peaks and zinc oxy-carbide complex structure peaks in our Raman and XPS spectra respectively. In XPS result, we observe the formation of Zn-C complex structure indicating the carbon substitution for oxygen sites. This augurs well with our GIXRD result. It can be observed from Table 1, that at the fluence of $1 \text{ pC}/\mu\text{m}^2$ the value of lattice parameter 'a' is slightly increases, while lattice parameter 'c' slightly decreases as compared to pristine sample. This increase in lattice parameter 'a' occurs due to the difference in their ionic radius of C^{4-} (0.260 nm) and O^{2-} (0.140 nm) [43,44]. The substitution of carbon for oxygen will expand the lattice. The presence of tensile stress in the film at the fluences of $1 \text{ pC}/\mu\text{m}^2$ also supports the enhancement of lattice parameter 'a' as well as substitution of carbon for oxygen. At the fluence of $10 \text{ pC}/\mu\text{m}^2$, the estimated value of stress is found to be negative, which signifies the presence of compressive stress and this reduces the lattice parameter 'a' at higher fluence. The Raman spectroscopy indicates the formation of oxygen vacancies, which is also confirmed by the XPS results. The carbon substituted at oxygen sites may lead to the formation of oxygen vacancies, Zn-C complex structures and tensile strain in the ZnO host matrix, which are responsible for the origin of ferromagnetism in the ZnO thin film.

5. Conclusions

In summary, we have observed room temperature ferromagnetism in 600 keV carbon ion irradiated ZnO thin films at different fluences of $1 \text{ pC}/\mu\text{m}^2$ and $10 \text{ pC}/\mu\text{m}^2$. A significant ferromagnetic ordering is achieved in ZnO thin films for $1 \text{ pC}/\mu\text{m}^2$ of fluence, and it decreases at higher fluence of $10 \text{ pC}/\mu\text{m}^2$ due to increased defect density and tensile strain in the system. The observed saturation magnetic moment is maximum for the film irradiated at fluence of $1 \text{ pC}/\mu\text{m}^2$. Our calculations show that each carbon ion produces an average magnetic moment of about $6.71 \mu_B$. GIXRD analysis confirms the decrement in the strain post irradiation, leading to tensile strain in the system. The Raman spectroscopy analysis established the presence of oxygen vacancies and tensile strain as well. XPS analysis also reveals the presence of oxygen vacancies as well as Zn-C complex structure. Thus, it is found that defects such as oxygen vacancy, formation of zinc-carbide complex structure and tensile strain are responsible for room temperature ferromagnetism in the present case. These experimental results clearly indicate that a specific band of defect, such as oxygen vacancies and Zn-C structures trigger the observed ferromagnetism. The carbon ion beam irradiation technique for inducing room temperature ferromagnetic ordering in ZnO opens a new direction for applications in spintronic devices.

CRedit authorship contribution statement

Sheshamani Singh: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. **Paras Poswal:** Investigation, Validation, Writing – review & editing. **Balakrishnan Sundaravel:** Investigation, Validation. **Sujay Chakravarty:** Formal analysis, Investigation, Resources, Validation. **Neeraj Shukla:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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